
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

Numerical Methods Laboratory Manual

Solution of Curve-Fitting Problems / Least Squares

Chapter 4 · Curve Fitting

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Topics: Fitting a Straight Line, a Parabola,
Multiple Linear, Nonlinear (Linearized), and
Weighted Least-Squares Curves

Reference: S.S. Sastry, *Introductory Methods of
Numerical Analysis*, 5th ed., PHI Learning — §4.1–4.3

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1 Introduction and How to Use This Manual

This manual accompanies the lecture material on **Chapter 4: Least-Squares Curve Fitting**, following S.S. Sastry, *Introductory Methods of Numerical Analysis* (§4.1–4.3). Each experiment treats one *curve-fitting procedure* and is organised uniformly:

- **Objective** — what the experiment teaches.
- **Theory** — a short statement of the model, its normal equations and (where needed) the linearizing transform.
- **Algorithm** — complete *pseudocode* you must implement.
- **Worked Example** — the book’s own numbered example(s), reproduced with the *exact input*, the *complete table of sums*, and the boxed fitted equation.
- **Exercise** — at least ten problems to be solved and reported.

Notation (Sastry). Throughout, m denotes the number of data points (x_i, y_i) , and the fitted straight line is written in the book’s form

$$Y = a_0 + a_1x,$$

i.e. a_0 is the intercept and a_1 the slope (*not* $a + bx$). The fit is obtained by minimising the sum of squared residuals $S = \sum_i (y_i - f(x_i))^2$; setting $\partial S / \partial (\text{each parameter}) = 0$ gives the *normal equations*. Report coefficients to **four decimal places** and tabulate *every* column of sums.

A general-purpose driver (in C/C++/Python/MATLAB) is recommended: read the data points, accumulate the required sums ($\sum x$, $\sum y$, $\sum x^2$, $\sum xy, \dots$ or their weighted counterparts), assemble and solve the normal equations, then print the fitted coefficients and a residual/fit-check row for each point.

2 Lab Report: Format and Submission Guidelines

Every submitted report **must** follow the structure and rules below.

2.1 Report structure

1. **Cover page** — Course title and code, Experiment number and name, your full name, roll/registration number, section, group, date of performance, date of submission.
2. **Objective** — one or two lines.
3. **Theory** — the model, its normal equations and linearizing transform (in your own words; 4–6 lines).
4. **Algorithm / Flowchart** — the pseudocode (or a flowchart) you implemented.
5. **Source code** — complete, commented program listing.
6. **Sample input** — the data points (x_i, y_i) you used.
7. **Output / Results** — the *full table of sums* and the fitted equation. Paste the program output *and* present a neat hand-formatted table with a residual/fit check.
8. **Discussion** — the values of the coefficients, the goodness of fit, the effect of the linearizing transform, and any difficulty faced.
9. **Conclusion** — 2–3 lines.

2.2 Formatting rules

- Paper size **A4**; margins **2.5 cm** on all sides.
- Body font **12 pt** serif (Times/Latin Modern); line spacing **1.5**; justified text.
- Headings bold; code in a **monospaced** font (e.g. Consolas/Courier), 10–11 pt, with line numbers.
- Every table and figure must be **numbered and captioned**. Tabulate the column of sums with aligned decimal points.
- Number all equations you reference.

2.3 Submission rules

- Submit a single **PDF** named `Roll_ExpNo_MethodName.pdf` — for example, `2303001_Exp01_StraightLineFit.pdf`.
- Submit the **source files** separately in a folder/zip with the same base name.
- Deadline: **within two week** of performing the experiment unless announced otherwise. Late reports lose marks as per course policy.
- Work must be **your own**. Copied reports/code receive zero; both provider and copier are penalised. You may discuss ideas but must write and code independently.

Marking (typical). Algorithm & code correctness 40%, results & tables of sums 25%, discussion 20%, formatting & neatness 10%, viva/demo 5%.

3 Experiment 1: Fitting a Straight Line, $Y = a_0 + a_1x$ (§4.2.1)

Objective

To fit a straight line $Y = a_0 + a_1x$ to a set of m data points (x_i, y_i) by the principle of least squares, by forming and solving the two normal equations, and to measure the goodness of fit through the correlation coefficient.

3.1 Theory

For the model $Y = a_0 + a_1x$, the sum of squared residuals is $S = \sum_{i=1}^m (y_i - a_0 - a_1x_i)^2$. Setting $\partial S / \partial a_0 = 0$ and $\partial S / \partial a_1 = 0$ gives the **normal equations**

$$m a_0 + a_1 \sum x = \sum y, \quad a_0 \sum x + a_1 \sum x^2 = \sum xy. \quad (1)$$

Solving (1) gives the slope and intercept

$$a_1 = \frac{m \sum xy - \sum x \sum y}{m \sum x^2 - (\sum x)^2}, \quad a_0 = \bar{y} - a_1 \bar{x}, \quad (2)$$

where $\bar{x} = \frac{1}{m} \sum x$ and $\bar{y} = \frac{1}{m} \sum y$. The last relation shows that *the least-squares line always passes through the centroid $(\bar{x}, \bar{y})$* . The quality of the fit is reported by the **correlation coefficient**

$$cc = \sqrt{\frac{S_T - S}{S_T}}, \quad S_T = \sum (y - \bar{y})^2, \quad S = \sum (y - a_0 - a_1x)^2, \quad (3)$$

with $cc \rightarrow 1$ for a near-perfect fit.

Algorithm 1 Fitting a Straight Line $Y = a_0 + a_1x$ by Least Squares

Input: data points (x_i, y_i) , $i = 1 \dots m$

Output: coefficients a_0, a_1 and correlation coefficient cc

```

1:  $S_x \leftarrow 0$ ;  $S_y \leftarrow 0$ ;  $S_{xx} \leftarrow 0$ ;  $S_{xy} \leftarrow 0$ 
2: for  $i = 1$  to  $m$  do
3:    $S_x \leftarrow S_x + x_i$ ;  $S_y \leftarrow S_y + y_i$ 
4:    $S_{xx} \leftarrow S_{xx} + x_i^2$ ;  $S_{xy} \leftarrow S_{xy} + x_i y_i$ 
5: end for
6:  $D \leftarrow m S_{xx} - S_x^2$  ▷ determinant of the normal system
7:  $a_1 \leftarrow (m S_{xy} - S_x S_y) / D$ 
8:  $\bar{x} \leftarrow S_x / m$ ;  $\bar{y} \leftarrow S_y / m$ 
9:  $a_0 \leftarrow \bar{y} - a_1 \bar{x}$ 
10:  $S_T \leftarrow \sum_i (y_i - \bar{y})^2$ ;  $S \leftarrow \sum_i (y_i - a_0 - a_1 x_i)^2$ 
11:  $cc \leftarrow \sqrt{(S_T - S) / S_T}$ 
12: return  $a_0, a_1, cc$ 

```

3.2 Worked Examples (full input and output)

Example 4.1 — straight line with correlation coefficient

Input. The five points

$$(1, 0.6), (2, 2.4), (3, 3.5), (4, 4.8), (5, 5.7).$$

Output.

x	y	x^2	xy	$(y - \bar{y})^2$	$(y - a_0 - a_1x)^2$
1	0.6	1	0.6	7.84	0.0784
2	2.4	4	4.8	1.00	0.0676
3	3.5	9	10.5	0.01	0.0100
4	4.8	16	19.2	1.96	0.0196
5	5.7	25	28.5	5.29	0.0484
$\sum x = 15$	$\sum y = 17.0$	$\sum x^2 = 55$	$\sum xy = 63.6$	$S_T = 16.10$	$S = 0.2240$

Here $m = 5$, $\bar{x} = 3$, $\bar{y} = 3.4$. From (2),

$$a_1 = \frac{5(63.6) - 15(17.0)}{5(55) - 15^2} = \frac{63}{50} = 1.26, \quad a_0 = \bar{y} - a_1\bar{x} = 3.4 - 1.26(3) = -0.38.$$

Hence

$$\boxed{Y = -0.38 + 1.26x}.$$

The correlation coefficient from (3) is

$$cc = \sqrt{\frac{16.10 - 0.2240}{16.10}} = \boxed{0.9930},$$

an excellent fit.

Example 4.2 — straight line from the normal equations

Input. The four points

$$(0, -1), (2, 5), (5, 12), (7, 20).$$

Output.

x	y	x^2	xy
0	-1	0	0
2	5	4	10
5	12	25	60
7	20	49	140
$\sum x = 14$	$\sum y = 36$	$\sum x^2 = 78$	$\sum xy = 210$

With $m = 4$ the normal equations (1) are

$$4a_0 + 14a_1 = 36, \quad 14a_0 + 78a_1 = 210.$$

Solving gives

$$\boxed{a_0 = -1.1381, \quad a_1 = 2.8966} \Rightarrow Y = -1.1381 + 2.8966x.$$

(This data set is revisited under *weighting* in Experiment 5.)

Exercise 1 (Straight Line $Y = a_0 + a_1x$)

Fit $Y = a_0 + a_1x$ by least squares; tabulate x , y , x^2 , xy (and, where asked, the residual columns), and report a_0, a_1 to 4 decimal places.

1. $(1, 0.6), (2, 2.4), (3, 3.5), (4, 4.8), (5, 5.7)$; also report cc .
2. $(0, -1), (2, 5), (5, 12), (7, 20)$.
3. $x : 1, 2, 3, 4, 5$; $y : 1.2, 2.0, 2.3, 3.1, 3.9$.
4. $x : 0, 1, 2, 3, 4$; $y : 1.0, 1.8, 3.3, 4.5, 6.3$.
5. $x : 10, 20, 30, 40, 50$; $y : 8, 13, 18, 24, 28$.
6. $x : 1, 3, 4, 6, 8, 9, 11, 14$; $y : 1, 2, 4, 4, 5, 7, 8, 9$.
7. $x : -2, -1, 0, 1, 2$; $y : -3.1, -0.9, 1.0, 3.2, 5.1$.
8. $x : 1, 2, 3, 4, 5, 6$; $y : 6.0, 4.8, 3.5, 2.6, 1.4, 0.2$ (a falling line).
9. **(Thermometer calibration)** A thermometer reading x ($^{\circ}\text{C}$) is checked against a standard y ($^{\circ}\text{C}$): $x : 20, 40, 60, 80, 100$; $y : 20.5, 40.3, 60.8, 80.2, 100.6$. Find the calibration line.
10. **(Ohm's law)** Currents $I = 1, 2, 3, 4, 5$ A give voltages $V = 3.1, 5.9, 9.2, 12.1, 14.9$ V. Fit $V = a_0 + a_1I$ and read off the resistance a_1 (Ω); then estimate V at $I = 6$ A.
11. **(Hooke's law)** A spring is loaded with masses $x : 0, 1, 2, 3, 4, 5$ (kg), giving extensions $y : 0.0, 2.1, 3.9, 6.2, 7.8, 10.1$ (cm). Fit $Y = a_0 + a_1x$ and report the stiffness a_1 .
12. $x : 100, 120, 140, 160, 180, 200$; $y : 0.45, 0.55, 0.60, 0.70, 0.80, 0.85$.

4 Experiment 2: Multiple Linear Least Squares, $z = a_0 + a_1x + a_2y$ (§4.2.2)

Objective

To fit a plane $z = a_0 + a_1x + a_2y$ to data (x_i, y_i, z_i) by least squares, by forming and solving the three normal equations in a_0, a_1, a_2 .

4.1 Theory

When the dependent variable z depends linearly on two independent variables x and y , we fit $z = a_0 + a_1x + a_2y$. Minimising $S = \sum (z_i - a_0 - a_1x_i - a_2y_i)^2$ with respect to a_0, a_1, a_2 gives the three **normal equations** (book eq. (4.11))

$$\begin{aligned} m a_0 + a_1 \sum x + a_2 \sum y &= \sum z, \\ a_0 \sum x + a_1 \sum x^2 + a_2 \sum xy &= \sum xz, \\ a_0 \sum y + a_1 \sum xy + a_2 \sum y^2 &= \sum yz. \end{aligned} \tag{4}$$

Solving this 3×3 system (by Gaussian elimination or Cramer's rule) yields the best-fit plane.

Algorithm 2 Multiple Linear Least Squares $z = a_0 + a_1x + a_2y$

Input: data points (x_i, y_i, z_i) , $i = 1 \dots m$

Output: coefficients a_0, a_1, a_2

- 1: accumulate $\sum x, \sum y, \sum z, \sum x^2, \sum xy, \sum y^2, \sum xz, \sum yz$
 - 2: **for** $i = 1$ **to** m **do**
 - 3: update each running sum from (x_i, y_i, z_i)
 - 4: **end for**
 - 5: assemble the 3×3 system (4)
 - 6: solve for (a_0, a_1, a_2) by Gaussian elimination / Cramer's rule
 - 7: **return** a_0, a_1, a_2
-

4.2 Worked Example (full input and output)

Example 4.3 — fit $z = a_0 + a_1x + a_2y$

Input. The five points

$$(x, y, z) = (0, 0, 2), (1, 1, 4), (2, 3, 3), (4, 2, 16), (6, 8, 8).$$

Output.

x	y	z	x^2	xy	zx	y^2	yz
0	0	2	0	0	0	0	0
1	1	4	1	1	4	1	4
2	3	3	4	6	6	9	9
4	2	16	16	8	64	4	32
6	8	8	36	48	48	64	64
13	14	33	57	63	122	78	109

With $m = 5$, the normal equations (4) are

$$5a_0 + 13a_1 + 14a_2 = 33,$$

$$13a_0 + 57a_1 + 63a_2 = 122,$$

$$14a_0 + 63a_1 + 78a_2 = 109.$$

Solving gives

$$\boxed{a_0 = 2, \quad a_1 = 5, \quad a_2 = -3} \Rightarrow z = 2 + 5x - 3y.$$

Exercise 2 (Multiple Linear Fit $z = a_0 + a_1x + a_2y$)

Fit $z = a_0 + a_1x + a_2y$ by least squares; tabulate all eight columns of sums and report a_0, a_1, a_2 .

1. $(0, 0, 2), (1, 1, 4), (2, 3, 3), (4, 2, 16), (6, 8, 8)$.
2. $(1, 1, 6), (2, 1, 9), (1, 2, 8), (2, 2, 11), (3, 3, 16)$.
3. $(0, 0, 1), (1, 0, 3), (0, 1, 4), (1, 1, 6), (2, 1, 8)$.
4. $(1, 2, 7), (2, 1, 8), (3, 3, 14), (4, 2, 15), (5, 4, 21)$.
5. $(0, 0, 5), (2, 1, 9), (4, 2, 13), (6, 3, 17), (8, 4, 21)$.
6. $(1, 0, 2), (2, 1, 5), (3, 2, 8), (4, 3, 11), (5, 4, 14)$.
7. $(2, 3, 11), (4, 1, 13), (6, 5, 27), (8, 2, 26), (10, 4, 38)$.
8. $(0, 1, 3), (1, 2, 7), (2, 3, 11), (3, 4, 15), (4, 5, 19)$.
9. $(1, 1, 3.1), (2, 2, 5.0), (3, 1, 6.2), (4, 2, 8.1), (5, 3, 9.9)$.
10. **(Yield model)** Crop yield z depends on rainfall x and fertiliser y :
 $(10, 2, 52), (12, 3, 61), (14, 2, 60), (16, 4, 73), (18, 3, 72)$. Fit $z = a_0 + a_1x + a_2y$
and interpret a_1, a_2 .
11. **(Heat output)** (x, y, z) with x = flow, y = pressure:
 $(1, 5, 16), (2, 4, 17), (3, 6, 24), (4, 5, 25), (5, 7, 32)$. Fit the plane.

5 Experiment 3: Linearization of Nonlinear Laws (§4.2.3)

Objective

To fit nonlinear laws (reciprocal, exponential, power, geometric) by *linearising* them into the straight-line form $Y = A_0 + A_1X$ through a suitable change of variable, and then applying the straight-line least-squares procedure of Experiment 1.

5.1 Theory

Many laws become linear in transformed variables X, Y . Once written as $Y = A_0 + A_1X$, they are fitted by the normal equations (1); the original constants are then recovered. The five standard transforms are:

Law	Linear form	Transform
(a) $y = ax + \frac{b}{x}$	$xy = ax^2 + b$	$Y = xy, X = x^2$ ($A_1 = a, A_0 = b$)
(b) $xy^a = b$	$a \log y + \log x = \log b$	$Y = \log y, X = \log x$
(c) $y = ab^x$	$\log y = \log a + x \log b$	$Y = \log y, X = x$
(d) $y = ax^b$	$\log y = \log a + b \log x$	$Y = \log y, X = \log x$
(e) $y = ae^{bx}$	$\ln y = \ln a + bx$	$Y = \ln y, X = x$

In each case fit $Y = A_0 + A_1X$, then recover the constants (e.g. in case (e) $A_1 = b, a = e^{A_0}$).

Algorithm 3 Fitting a Nonlinear Law by Linearization

Input: data (x_i, y_i) and the chosen transform $X = g(x, y), Y = h(x, y)$

Output: original constants of the nonlinear law

- 1: **for** $i = 1$ **to** m **do**
- 2: $X_i \leftarrow g(x_i, y_i); Y_i \leftarrow h(x_i, y_i)$
- 3: **end for**
- 4: fit the line $Y = A_0 + A_1X$ by (1)–(2)
- 5: recover the original constants from A_0, A_1 (e.g. $a = e^{A_0}$ or 10^{A_0})
- 6: **return** original constants

5.2 Worked Examples (full input and output)

Example 4.4 — exponential law $y = a e^{bx}$ (transform e)

Input. The five points

$$(1, 2.473), (3, 6.722), (5, 18.274), (7, 49.673), (9, 135.026).$$

Use $X = x$, $Y = \ln y$, so $Y = A_0 + A_1 X$ with $A_1 = b$ and $a = e^{A_0}$.

Output.

$X = x$	$Y = \ln y$	X^2	XY
1	0.905	1	0.905
3	1.905	9	5.715
5	2.905	25	14.525
7	3.905	49	27.335
9	4.905	81	44.145
$\sum X = 25$	$\sum Y = 14.525$	$\sum X^2 = 165$	$\sum XY = 92.625$

Here $\bar{X} = 5$, $\bar{Y} = 2.905$. From (2),

$$A_1 = b = \frac{5(92.625) - 25(14.525)}{5(165) - 25^2} = 0.5, \quad A_0 = \bar{Y} - A_1 \bar{X} = 2.905 - 0.5(5) = 0.405.$$

Then $a = e^{0.405} = 1.499$, hence

$$y = 1.499 e^{0.5x}.$$

Example 4.5 — reciprocal law $y = x/(a + bx)$ **Input.** The four points

$$(3, 7.148), (5, 10.231), (8, 13.509), (12, 16.434).$$

Writing $\frac{1}{y} = b + a\frac{1}{x}$, put $Y = \frac{1}{y}$, $X = \frac{1}{x}$, so $Y = A_0 + A_1X$ with $A_0 = b$ and $A_1 = a$.

Output.

$X = 1/x$	$Y = 1/y$	X^2	XY
0.333	0.140	0.111	0.047
0.200	0.098	0.040	0.020
0.125	0.074	0.016	0.009
0.083	0.061	0.007	0.005
$\sum X = 0.741$	$\sum Y = 0.373$	$\sum X^2 = 0.174$	$\sum XY = 0.081$

Solving the line normal equations for these sums gives

$$A_1 = a = 0.324, \quad A_0 = b = 0.0331,$$

hence

$$y = \frac{x}{0.324 + 0.0331x}.$$

Remark. The data were generated from the true relation $y = \frac{x}{0.3162 + 0.0345x}$; the small differences arise from rounding the transformed columns to three decimals, illustrating that linearization fits the *transformed* residuals, not the original ones.

Exercise 3 (Nonlinear Laws by Linearization)

Choose the correct transform, fit $Y = A_0 + A_1 X$, and recover the original constants. Tabulate the X , Y , X^2 , XY columns.

1. ($y = ae^{bx}$) (1, 2.473), (3, 6.722), (5, 18.274), (7, 49.673), (9, 135.026).
2. ($y = x/(a + bx)$) (3, 7.148), (5, 10.231), (8, 13.509), (12, 16.434).
3. ($y = ae^{bx}$) $x : 0, 1, 2, 3, 4$; $y : 1.0, 1.65, 2.72, 4.48, 7.39$.
4. ($y = ax^b$, power) $x : 1, 2, 3, 4, 5$; $y : 2.0, 5.7, 10.4, 16.0, 22.4$.
5. ($y = ab^x$, geometric) $x : 1, 2, 3, 4, 5, 6$; $y : 5, 9, 16.2, 29.2, 52.5, 94.5$.
6. ($y = ax + b/x$) $x : 1, 2, 3, 4, 5$; $y : 5.1, 5.6, 7.0, 8.8, 10.6$ (fit $xy = ax^2 + b$).
7. ($xy^a = b$) $x : 1, 2, 4, 8$; $y : 8.0, 5.66, 4.0, 2.83$.
8. **(Capacitor discharge)** At $t = 0, 1, 2, 3, 4, 5$ s the voltage is $V = 10, 6.1, 3.7, 2.2, 1.35, 0.82$ V. Fit $V = V_0 e^{-t/\tau}$ and report V_0, τ .
9. **(Bacterial growth)** At $t = 0, 1, 2, 3, 4$ h the count is $N = 100, 150, 228, 340, 510$. Fit $N = N_0 e^{bt}$ and find the doubling time $\ln 2/b$.
10. **(Allometry)** Body mass $m = 1, 2, 3, 5, 8$ (kg) versus metabolic rate $r = 70, 118, 160, 230, 330$. Fit $r = am^b$ and check $b \approx 0.75$.
11. ($y = ax^b$) $x : 1, 2, 4, 6, 8, 10$; $y : 1.0, 2.83, 8.0, 14.7, 22.6, 31.6$.

6 Experiment 4: Curve Fitting by a Polynomial / Parabola (§4.2.4)

Objective

To fit a polynomial $Y = a_0 + a_1x + \cdots + a_nx^n$ — and in particular a second-degree parabola ($n = 2$) — to data by least squares, by forming and solving the $(n+1)$ normal equations.

6.1 Theory

For the degree- n polynomial $Y = a_0 + a_1x + \cdots + a_nx^n$, minimising $S = \sum (y_i - a_0 - a_1x_i - \cdots - a_nx_i^n)^2$ gives $n+1$ normal equations. For a **parabola** ($n = 2$), $Y = a_0 + a_1x + a_2x^2$, the three normal equations are

$$\begin{aligned} m a_0 + a_1 \sum x + a_2 \sum x^2 &= \sum y, \\ a_0 \sum x + a_1 \sum x^2 + a_2 \sum x^3 &= \sum xy, \\ a_0 \sum x^2 + a_1 \sum x^3 + a_2 \sum x^4 &= \sum x^2y. \end{aligned} \quad (5)$$

They require the sums $\sum x, \sum x^2, \sum x^3, \sum x^4$ and $\sum y, \sum xy, \sum x^2y$.

Algorithm 4 Fitting a Parabola $Y = a_0 + a_1x + a_2x^2$ by Least Squares

Input: data points (x_i, y_i) , $i = 1 \dots m$

Output: coefficients a_0, a_1, a_2

- 1: accumulate $\sum x, \sum x^2, \sum x^3, \sum x^4$ and $\sum y, \sum xy, \sum x^2y$
- 2: **for** $i = 1$ **to** m **do**
- 3: update each running sum from (x_i, y_i)
- 4: **end for**
- 5: assemble the 3×3 system (5)
- 6: solve for (a_0, a_1, a_2) by Gaussian elimination / Cramer's rule
- 7: **return** a_0, a_1, a_2

6.2 Worked Examples (full input and output)

Example 4.6 — parabola through three points

Input. The three points $(0, 1), (1, 6), (2, 17)$.

Output.

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	6	1	1	1	6	6
2	17	4	8	16	34	68
3	24	5	9	17	40	74

With $m = 3$, the normal equations (5) are

$$3a_0 + 3a_1 + 5a_2 = 24, \quad 3a_0 + 5a_1 + 9a_2 = 40, \quad 5a_0 + 9a_1 + 17a_2 = 74.$$

Solving gives

$$\boxed{a_0 = 1, \quad a_1 = 2, \quad a_2 = 3} \Rightarrow Y = 1 + 2x + 3x^2 \text{ (exact).}$$

Example 4.7 — least-squares parabola**Input.** The four points

$$(1, 0.63), (3, 2.05), (4, 4.08), (6, 10.78).$$

Output.

x	y	x^2	x^3	x^4	xy	x^2y
1	0.63	1	1	1	0.63	0.63
3	2.05	9	27	81	6.15	18.45
4	4.08	16	64	256	16.32	65.28
6	10.78	36	216	1296	64.68	388.08
14	17.54	62	308	1634	87.78	472.44

With $m = 4$, the normal equations (5) are

$$\begin{aligned} 4a_0 + 14a_1 + 62a_2 &= 17.54, \\ 14a_0 + 62a_1 + 308a_2 &= 87.78, \\ 62a_0 + 308a_1 + 1634a_2 &= 472.44. \end{aligned}$$

Solving gives

$$a_0 = 1.24, \quad a_1 = -1.05, \quad a_2 = 0.44 \Rightarrow Y = 1.24 - 1.05x + 0.44x^2.$$

Exercise 4 (Polynomial / Parabola)Fit $Y = a_0 + a_1x + a_2x^2$ (or a cubic where indicated) by least squares; tabulate all columns of sums and report the coefficients.

1. $(0, 1), (1, 6), (2, 17)$.
2. $(1, 0.63), (3, 2.05), (4, 4.08), (6, 10.78)$.
3. $x : 0, 1, 2, 3, 4; y : 1, 1.8, 1.3, 2.5, 6.3$.
4. $x : 1, 2, 3, 4, 5, 6; y : 2.0, 5.0, 10.0, 17.0, 26.0, 37.0$ (a perfect x^2+1).
5. $x : -2, -1, 0, 1, 2; y : 4.1, 1.0, 0.1, 1.1, 3.9$.
6. $x : 0, 2, 4, 6, 8, 10; y : 5.0, 6.2, 9.0, 12.5, 18.0, 24.5$.
7. $x : 1, 2, 3, 4, 5, 6, 7; y : 2.3, 5.2, 9.7, 16.5, 24.5, 34.0, 46.0$.
8. $x : 1, 1.5, 2, 2.5, 3, 3.5, 4; y : 1.1, 1.3, 1.6, 2.0, 2.7, 3.4, 4.1$.
9. $x : 10, 15, 20, 25, 30; y : 35.3, 32.4, 29.2, 28.1, 31.0$.
10. **(Projectile motion)** Heights h at times t : $t : 0, 1, 2, 3, 4$ s; $h : 2, 27, 42, 47, 42$ m. Fit $h = a_0 + a_1t + a_2t^2$ and confirm $h = 2 + 30t - 5t^2$; then find the time and value of maximum height.
11. **(Cubic fit)** $x : 0, 1, 2, 3, 4, 5; y : 1, 0, 1, 10, 33, 76$. Fit $Y = a_0 + a_1x + a_2x^2 + a_3x^3$.

7 Experiment 5: Weighted Least Squares (§4.3)

Objective

To fit a straight line $Y = a_0 + a_1x$ by **weighted** least squares, in which each data point carries a positive weight W_i reflecting its reliability, and to observe how increasing the weight of a trusted point pulls the fitted line toward it.

7.1 Theory

When some observations are more reliable than others, we minimise the *weighted* sum of squared residuals

$$S = \sum_i W_i [y_i - f(x_i)]^2, \quad W_i > 0. \quad (6)$$

Taking all $W_i = 1$ recovers ordinary least squares. For the line $f(x) = a_0 + a_1x$, setting $\partial S/\partial a_0 = 0$ and $\partial S/\partial a_1 = 0$ gives the **weighted normal equations** (book eqs. (4.31)–(4.32))

$$a_0 \sum W + a_1 \sum Wx = \sum Wy, \quad a_0 \sum Wx + a_1 \sum Wx^2 = \sum Wxy. \quad (7)$$

The required accumulations are the weighted sums $\sum W$, $\sum Wx$, $\sum Wx^2$, $\sum Wy$, $\sum Wxy$. A *weighted polynomial* fit (§4.3.2) follows the same idea: each normal equation of (5) is formed from weighted sums $\sum Wx^k$ and $\sum Wx^ky$.

Algorithm 5 Weighted Least-Squares Line $Y = a_0 + a_1x$

Input: data (x_i, y_i) with weights $W_i > 0$, $i = 1 \dots m$

Output: coefficients a_0, a_1

1: $SW \leftarrow 0$; $SWx \leftarrow 0$; $SWx2 \leftarrow 0$; $SWy \leftarrow 0$; $SWxy \leftarrow 0$

2: **for** $i = 1$ **to** m **do**

3: $SW \mathrel{+}= W_i$; $SWx \mathrel{+}= W_i x_i$; $SWx2 \mathrel{+}= W_i x_i^2$

4: $SWy \mathrel{+}= W_i y_i$; $SWxy \mathrel{+}= W_i x_i y_i$

5: **end for**

6: solve $\begin{bmatrix} SW & SWx \\ SWx & SWx2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} SWy \\ SWxy \end{bmatrix}$

7: **return** a_0, a_1

7.2 Worked Examples (full input and output)

Example 4.9 — weighting the reliable point ($W = 10$)

Input. The data of Example 4.2, $(0, -1), (2, 5), (5, 12), (7, 20)$, where the point $(5, 12)$ is the most reliable and is given weight $W = 10$; the others have $W = 1$.

Output.

x	y	W	Wx	Wx^2	Wy	Wxy
0	-1	1	0	0	-1	0
2	5	1	2	4	5	10
5	12	10	50	250	120	600
7	20	1	7	49	20	140
$\sum W = 13 \quad \sum Wx = 59 \quad \sum Wx^2 = 303 \quad \sum Wy = 144 \quad \sum Wxy = 750$						

The weighted normal equations (7) are

$$13a_0 + 59a_1 = 144, \quad 59a_0 + 303a_1 = 750.$$

Solving gives

$$a_0 = -1.3493, \quad a_1 = 2.7380 \Rightarrow y = -1.3493 + 2.7380x.$$

Example 4.10 — heavier weighting ($W = 100$)

Input. The same data, now with weight $W = 100$ on $(5, 12)$ and $W = 1$ elsewhere.

Output. The weighted sums become

$$\sum W = 103, \quad \sum Wx = 509, \quad \sum Wx^2 = 2553, \quad \sum Wy = 1224, \quad \sum Wxy = 6150,$$

so the weighted normal equations (7) are

$$103a_0 + 509a_1 = 1224, \quad 509a_0 + 2553a_1 = 6150.$$

Solving gives

$$a_0 = -1.4126, \quad a_1 = 2.6906 \Rightarrow y = -1.4126 + 2.6906x.$$

At the trusted point, the fitted value is $y(5) = -1.4126 + 2.6906(5) = 12.0402$, much closer to the observed 12 than the unweighted fit of Example 4.2.

Observation. As the weight on $(5, 12)$ grows ($W = 1 \rightarrow 10 \rightarrow 100$), the line is pulled toward that point and $y(5) \rightarrow 12$: the fit at the trusted observation improves monotonically with its weight. The same weighting principle extends to polynomials (§4.3.2) by forming each normal equation from the weighted sums $\sum Wx^k$ and $\sum Wx^ky$.

Exercise 5 (Weighted Least Squares)

Fit $Y = a_0 + a_1x$ by weighted least squares; tabulate the columns W , Wx , Wx^2 , Wy , Wxy and report a_0, a_1 . Compare with the unweighted fit.

1. $(0, -1), (2, 5), (5, 12), (7, 20)$ with $W = 10$ on $(5, 12)$, else $W = 1$.
2. The same data with $W = 100$ on $(5, 12)$; compare $y(5)$ with Exercise 1.
3. The same data with $W = 1$ everywhere (recover Example 4.2).
4. $x : 1, 2, 3, 4, 5$; $y : 1.2, 2.0, 2.3, 3.1, 3.9$ with $W : 1, 1, 1, 1, 5$.
5. $x : 0, 1, 2, 3, 4$; $y : 1.0, 1.8, 3.3, 4.5, 6.3$ with $W : 2, 1, 1, 1, 2$.
6. $x : 10, 20, 30, 40, 50$; $y : 8, 13, 18, 24, 28$ with $W : 1, 1, 3, 1, 1$.
7. $x : 1, 2, 3, 4, 5, 6$; $y : 6.0, 4.8, 3.5, 2.6, 1.4, 0.2$ with $W : 1, 2, 3, 3, 2, 1$ (trust the middle).
8. $x : -2, -1, 0, 1, 2$; $y : -3.1, -0.9, 1.0, 3.2, 5.1$ with $W : 1, 1, 4, 1, 1$.
9. $x : 0, 5, 10, 15, 20$; $y : 0.32, 0.29, 0.26, 0.23, 0.20$ with $W : 5, 1, 1, 1, 5$ (anchor the ends).
10. **(Calibration with a precise endpoint)** $x : 20, 40, 60, 80, 100$; $y : 20.5, 40.3, 60.8, 80.2, 100.6$ with $W = 20$ on the certified point $(100, 100.6)$, else $W = 1$.
11. For Exercise 5.4–5.5, repeat with all weights $= 1$ and tabulate the change in a_0, a_1 caused by the weighting.

Appendix: Quick Comparison of the Procedures

Procedure	Model	Normal equations / transform
Straight line (§4.2.1)	$Y = a_0 + a_1x$	$m a_0 + a_1 \sum x = \sum y; a_0 \sum x + a_1 \sum x^2 = \sum xy$
Multiple linear (§4.2.2)	$z = a_0 + a_1x + a_2y$	three eqns (4.11) in $\sum x, \sum y, \sum x^2, \sum xy, \sum y^2, \sum xz, \sum yz$
Nonlinear (§4.2.3)	$ae^{bx}, ax^b, ab^x,$ $ax + b/x$	linearize to $Y = A_0 + A_1X$ (log / reciprocal), then fit the line
Polynomial / parabola (§4.2.4)	$Y =$ $a_0 + a_1x + a_2x^2$	three eqns using $\sum x, \sum x^2, \sum x^3, \sum x^4, \sum y, \sum xy, \sum x^2y$
Weighted (§4.3)	$Y = a_0 + a_1x,$ $W_i > 0$	$a_0 \sum W + a_1 \sum Wx = \sum Wy;$ $a_0 \sum Wx + a_1 \sum Wx^2 = \sum Wxy$

Reference: S.S. Sastry, *Introductory Methods of Numerical Analysis*, 5th ed., PHI Learning — §4.1–4.3.